

PAVANI RAJKONKIMALLA

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PROFESSIONAL SUMMARY

- Senior Data Engineer with 10+ years of IT experience specializing in Azure Cloud, ETL/ELT pipeline development, data warehouse engineering, and large-scale distributed data processing.
- Proven expertise in designing and building scalable batch and real-time data pipelines using Azure Data Factory, Azure Databricks, PySpark, Apache Spark, and Snowflake.
- Hands-on experience with SQL, Python, and Spark for production-grade data solutions. Skilled in Star Schema and Snowflake Schema data modeling, Data Vault architecture, CI/CD, DevOps practices, and Agile methodologies.
- Deep expertise in Azure Data Lake Gen2 (ADLS Gen2), Azure Event Hubs, Azure DevOps, Azure Blob Storage, data security (RBAC, data masking), and data quality frameworks across TB-scale datasets.
- Hands-on experience with Microsoft Fabric (very trending in 2024–2025) for unified analytics, OneLake storage, and Fabric pipelines; proficient in Azure Synapse Analytics for large-scale data warehousing and big data analytics.
- Proficient in dbt (data build tool) for building modular, tested, and version-controlled transformation layers on top of Snowflake and other cloud warehouses.
- Experienced with Apache Airflow for end-to-end pipeline orchestration, DAG management, scheduling, and monitoring across complex multi-environment data workflows.
- Strong knowledge of Medallion Architecture (Bronze/Silver/Gold) for structured data lake design, enabling reliable, layered data ingestion from raw to curated zones.
- Expertise in Dimensional Modeling (Kimball), Data Vault Modeling, OLAP, and Data Mart design for scalable enterprise data warehouse solutions.
- Experienced in containerizing data workloads using Docker and orchestrating deployments on Kubernetes / Azure Kubernetes Service (AKS) for high-availability, scalable data processing.
- Skilled in Infrastructure as Code using Terraform and Bicep for automated, repeatable cloud environment provisioning across Dev, UAT, and Production.
- Proficient in enterprise data governance using Microsoft Purview for data cataloging, lineage tracking, sensitivity classification, and compliance policy enforcement.
- Expertise in Data Quality Monitoring frameworks to enforce data integrity, completeness, and freshness SLAs across TB-scale datasets in production pipelines.
- Experienced in real-time and event-driven architectures using Apache Kafka, Azure Event Hubs, and Azure Stream Analytics for high-throughput transaction and streaming data processing.
- Strong background in Azure SQL Database, Azure SQL Data Warehouse, Snowflake, and NoSQL systems with expertise in performance tuning, stored procedures, and query optimization on large-scale datasets.
- Adept at building PowerBI dashboards, REST API integrations, and working across data formats including Parquet, ORC, Avro, JSON, and CSV in cloud-native pipelines.

TECHNICAL SKILLS

- **Programming Languages:** Python (OOP, packaging, logging, Pandas), PySpark (DataFrames, Spark SQL, optimization), SQL/T-SQL (window functions, CTEs, performance tuning), Scala, Java
- **Data Storage & Warehousing:** Azure Data Lake Storage Gen2 (ADLS Gen2), Azure Synapse Analytics, Azure SQL Database, Snowflake, SnowSQL, Snowpipe, Microsoft Fabric, SQL Server, NoSQL
- **Data Integration & ETL/ELT:** Azure Data Factory (v2), Azure Databricks, Apache Airflow, dbt (data build tool), SSIS, Delta Lake, batch & streaming pipelines, incremental loads, data validation & quality frameworks
- **Big Data & Distributed Systems:** Apache Spark (Core, SQL, Streaming), Hadoop (HDFS, YARN, MapReduce), Apache Kafka, Hive, Apache Ozone, data partitioning & bucketing
- **Streaming & Real-Time Data:** Azure Event Hubs, Azure Stream Analytics, Apache Kafka, Azure Service Bus, Azure Logic Apps
- **DevOps & DataOps:** Azure DevOps, CI/CD pipelines, Git, Infrastructure as Code (Terraform, Bicep), Docker, Kubernetes, Azure Kubernetes Service (AKS), unit testing, integration testing, Agile/Scrum
- **Data Modeling & Governance:** Medallion Architecture (Bronze/Silver/Gold), Dimensional Modeling (Kimball), Data Vault Modeling, Star Schema, Snowflake Schema, OLAP, Data Marts, Microsoft Purview, Data Quality Monitoring
- **Cloud – Azure:** Azure Data Factory, Azure Databricks, ADLS Gen2, Azure Blob Storage, Azure Synapse Analytics, Azure SQL DB/DW, Azure Event Hubs, Azure Key Vault, Azure DevOps, Microsoft Fabric
- **Data Formats & Reporting:** Parquet, ORC, Avro, JSON, XML, CSV; PowerBI, SSRS, REST APIs

- **Security & Compliance:** RBAC, data masking, Azure Key Vault, table-level security in Databricks, Microsoft Purview

PROFESSIONAL EXPERIENCE

UBC

King of Prussia, Pennsylvania

Senior Data Warehouse Engineer

May 2023 – Feb 2026

- Involved in Requirement gathering, business Analysis, Design and Development, testing and implementation of business rules.
- Understand business use cases, integration business, write business & technical requirements documents, logic diagrams, process flow charts, and other application related documents.
- Built a common SFTP download/upload framework using Azure Data Factory and Snowflake.
- Involved in logical modeling, physical database design, data sourcing and data transformation, data loading, SQL and performance tuning.
- Worked on Azure Cloud services and Snowflake development.
- Loaded the tables from the Azure Data Lake to Azure Blob Storage for pushing them to Snowflake.
- Developing ADF Pipelines in and out of data warehouse, develop major regulatory and financial reports using advanced SQL queries in Snowflake.
- Design and develop ETL pipelines using ADF for data ingestion and transformation.
- Collaborate with Azure stack modules like Data Lakes and SQL DW to build robust data solutions.
- Write SQL, Python, and PySpark code for efficient data processing and transformation.
- Understand and translate business requirements into technical designs.
- Develop mapping documents and transformation rules as per project scope.
- Communicate project status with stakeholders, ensuring smooth project execution.

Environment: Azure Data Factory (v2), Azure Data Lake Gen2 (ADLS Gen2), Azure Databricks, Azure Synapse Analytics, Snowflake, Microsoft Fabric, dbt, Apache Airflow, Azure Blob Storage, Azure DevOps, Terraform, Bicep, Docker, Kubernetes (AKS), Microsoft Purview, SQL Server, Python, PySpark

Banner Bank

Walla Walla, WA

Data Engineer

May 2021 – Feb 2023

- Designed and maintained batch and streaming ETL/ELT pipelines using Azure Data Factory and Azure Databricks, implementing Medallion Architecture (Bronze/Silver/Gold) to process structured, semi-structured, and unstructured financial data at scale.
- Developed PySpark applications in Databricks for data extraction, transformation, and aggregation across multiple file formats (Parquet, ORC, Avro, JSON, CSV); built dbt models for modular, reusable SQL transformations in Snowflake.
- Implemented Delta Lake tables in Azure Databricks with table-level RBAC security; applied schema design and data validation checks to ensure data integrity and compliance with financial regulations.
- Built bulk loading workflows from Azure Data Lake Gen2 (ADLS Gen2) to Snowflake cloud; authored complex SnowSQL scripts for business analysis and financial reporting dashboards.
- Engineered Azure Stream Analytics solutions using Azure Event Hubs and Apache Kafka for real-time transaction monitoring; implemented approval workflows via Azure Logic Apps and Service Bus.
- Orchestrated data pipelines using Apache Airflow for scheduling and dependency management across multi-environment deployments (Dev, UAT, Production).
- Containerized data processing workloads using Docker and deployed on Azure Kubernetes Service (AKS) to ensure scalability and resource efficiency for large-scale batch jobs.
- Implemented CI/CD pipelines using Azure DevOps and Git; provisioned cloud infrastructure using Terraform/Bicep as Infrastructure as Code for consistent environment management.
- Registered and managed data assets in Microsoft Purview for data governance, lineage tracking, and policy enforcement across Snowflake and Azure platforms.
- Developed data dashboards and visualization reports in PowerBI for business stakeholders; collaborated with data scientists and analysts to provide clean, reliable datasets following Medallion architecture principles.

Environment: Azure Data Factory (v2), Azure Data Lake Gen2 (ADLS Gen2), Azure Databricks, Azure Synapse Analytics, Snowflake, Microsoft Fabric, dbt, Apache Airflow, Apache Kafka, Azure Event Hubs, Docker, Kubernetes (AKS), Terraform, Microsoft Purview, Azure Blob Storage, Azure DevOps, SQL Server, PowerBI

Ameren

Missouri

Data Engineer

Nov 2019 – Apr 2021

- Led end-to-end migration of on-premises ETL processes to Azure Data Lake Gen2, designing scalable cloud-native data pipelines using Azure Data Factory (v2).
- Built ADF pipelines with Linked Services, Datasets, and Activities (ForEach, Lookup, Until, Web, Wait, If Condition) to extract, transform, and load data from Azure SQL, Blob Storage, SQLDW, and Data Lake.
- Implemented CI/CD pipelines using Azure DevOps and Git for automated deployment across Dev, SIT, UAT, and Production; resolved code defects during SIT and UAT testing cycles.
- Configured ADF Triggers and scheduled pipelines; built a custom alerts and monitoring platform to notify stakeholders of pipeline failures and data load SLA breaches.
- Used Azure Key Vault as a centralized secret repository, referenced securely in ADF pipelines; implemented Stored Procedure activity for metadata management.
- Wrote complex SQL queries, stored procedures, joins, functions, and packages for data validation and testing; performed SQL performance tuning on large-scale datasets.

Environment: Azure Data Factory (v2), Azure Data Lake Gen2, Azure Blob Storage, SQL Server, Logic Apps, Azure DevOps, Git

VISA

TX

Data Engineer

Sep 2017 – Nov 2019

- Designed and implemented large-scale data processing pipelines in Azure Databricks and PySpark to handle petabytes of transaction data from terabytes of client data sources.
- Built a reusable framework to convert fixed-width files to Parquet format using Azure Data Factory and Azure Databricks; automated daily file ingestion to Azure Data Lake.
- Designed and implemented a data validation framework supporting both RDBMS and file-based systems to identify invalid and duplicate records among billions of records.
- Developed a Delta ingestion framework using Azure Databricks and ADF to pull incremental data into Azure Data Lake; created Delta Lake tables with table-level RBAC security.
- Performed data transformations including sort, join, aggregations, and filter operations using PySpark DataFrames and Spark SQL to generate analytical datasets.
- Implemented approval workflows and email notifications using Azure Logic Apps; developed a PowerBI refresh framework from Azure Data Lake using Azure Databricks.
- Worked in Agile/Scrum to design and deliver primary provider identification solutions using Azure Databricks and PySpark.

Environment: Azure Databricks, Azure Data Factory, Azure Data Lake, Azure SQL, PySpark, Azure Key Vault, PowerBI, Azure DevOps

Ecolab Digital Center

India

BI Developer / Data Engineer

Apr 2010 – Jun 2014

- Designed and developed SSIS packages using Control Flow and Data Flow components (ETL) to extract, transform, and load customer data from Excel to SQL Server; ensured data quality and case-sensitivity validation using Python and SSIS.
- Built stored procedures, functions, and T-SQL scripts for customer data display and reporting; implemented retry framework for SSIS job recovery and automated email notifications post data load.
- Generated enterprise reports using SSRS from SQL Server (OLTP) including drill-downs, drill-throughs, sub-reports, and parameterized reports scheduled on daily/weekly/monthly cadence.
- Performed SQL performance tuning and database optimization; used DTS, SSIS, and BULK INSERT for large-volume data transfers between SQL Server databases.

Environment: SQL Server Management Studio, SSRS, SSIS, T-SQL, Python

EDUCATION

Jawaharlal Nehru Technological University (JNTU), India

Bachelor of Technology, Electronics and Communications Engineering

2006